Optimal Maneuver in Directional Naval Combat: Calibrated Firepower Kernels, Committed-Maneuver Games, and Torpedo Denial in an Auditable World War II Wargame

Research Layer of the *Iron Bottom Sound* project*

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Abstract

Classic surface-fleet tactics—crossing the T, local fire concentration, and torpedo threat—are usually justified by doctrine rather than by game-theoretic analysis. We develop a two-layer framework that makes these tactics objects of measurement. Layer 1 is a directional *firepower kernel*, a closed-form anisotropic surrogate of a ship's expected gunnery output, calibrated against the exact probabilistic adjudication of a fully auditable, deterministic WWII surface-combat rules engine (the open *Iron Bottom Sound* implementation): on held-out geometry grids the kernel reproduces engine expected hits with pooled Spearman $\rho = 0.991$ and normalized MAE 5.7% across four ship classes (DD/CL/CA/BB). Layer 2 embeds the kernel in a two-player differential game over the reduced relative state (r, α_B, α_R). We first prove—and verify numerically to machine precision—a *degeneracy proposition*: with simultaneous moves, identical instantaneous capabilities, and full observability, the stationary value collapses to the instantaneous fire differential, so positional maneuvering has zero equilibrium value. Maneuvering acquires value only under order commitment; we therefore solve the committed-maneuver (open-loop) minimax over a library of sealed battle plans and map tactical regimes over speed and gun-range ratios. Speed advantage is found to act through the reachable tactical set alone: at equal gun range the faster committed duelist is systematically *worse* off under a pure fire-differential objective, and speed acquires value only in combination with a range advantage. Counterfactual torpedo analysis implements the best-response definition of maneuver denial. In the low-direct-hit regime the denial value is exactly zero—with a measured mechanism: the position-value landscape is tiered with 25 tied optimal routes on average, so thin threats are absorbed by free lateral moves—whereas threats that saturate the entire top tier produce significant denial (0.92 value points, 95% CI $[0.42, 1.42]$). Hypothesis refinement, not confirmation, is the output. Finally, we validate the framework inside the rules engine

*All experiments in this paper are fully reproducible from the accompanying repository: every figure and number is generated by a script in `research/experiments/` against a frozen, deterministic rules engine. The study analyzes a historical board-wargame rules system; it does not constitute a real-world military decision aid.

itself: a receding-horizon geometry policy converted into legal engine orders beats a non-maneuvering baseline by a large, statistically significant paired damage differential in every geometry, is at parity with mature doctrine heuristics in two of three geometries, and concedes head-on to the line-ahead specialist—exactly the shape the degeneracy proposition predicts. A pre-specified analysis of geometry-induced engagement networks passes both acceptance gates under the plan's own metric definitions ($\Delta R^2 = +0.108$, MAE -17.5%); we also report the stricter specification in which it fails ($\Delta R^2 = +0.033$), localizing the predictive value to the directional in-degree itself. All experiments are seeded, paired, and reproducible from committed code.

1 Introduction

Naval officers have understood directional firepower for more than a century: Bernotti's fundamental tactics already treat the crossing of the enemy's line as an ideal to be weighed against the opponent's active resistance [1]. Yet in the modelling literature the geometric content of surface combat is usually dispersed across separate traditions: Lanchester-type attrition laws [3–5], networked fire models [6], pursuit–evasion and ship differential games [7–9], and empirical game-theoretic analysis of multi-agent populations [11, 13, 14]. What is missing is a single auditable pipeline in which the same directional-fire physics both (i) drives a tractable game-theoretic model whose tactical regimes can be characterized, and (ii) is validated against a complete, disaggregated rules engine of a real historical wargame.

This paper builds such a pipeline on top of the open *Iron Bottom Sound* (IBS) rules engine, a deterministic, command–event–state implementation of a WWII surface-combat board wargame with fully structured adjudication tables (gunnery hit and result tables, armor penetration, special damage, torpedo mechanics). The engine's adjudication is a pure function of (state, sealed orders, seed): identical seeds and orders reproduce byte-identical event streams, and every die roll is recorded with its rule reference. This determinism is exactly what a scientific study of tactics requires, and we make it a first-class experimental object (Section 3).

We ask four questions:

1. **RQ1 (kernel).** Can a smooth anisotropic firepower kernel calibrated *only* against the engine's own adjudication pipeline replace the discrete rule tables inside continuous optimization, with quantified fidelity?

2. **RQ2 (game).** Under what speed- and range-asymmetries does classical maneuvering (crossing the T, refusing battle, oblique duels) emerge as the game-theoretically optimal committed plan, and what is the value of speed advantage?

3. **RQ3 (denial).** Is the value of a torpedo salvo separable into direct damage and maneuver denial, where denial is defined strictly by the target's best-response change rather than by hand-labelled tactics?

4. **RQ4 (validation).** Does a policy derived from the continuous model translate into legal engine orders and survive contact with the full rules engine against doctrine baselines?

Our methodological stance is deliberately conservative and is enforced by the research layer's discipline: no rule table is ever copied into research code (the kernel queries engine functions); no hidden information leaks into policies (audited); Crossing-the-T and formation-splitting are never rewarded (the payoff is expected-hits differential only); and every comparison uses paired seeds with confidence intervals. Where a pre-specified acceptance threshold is not met we report the null result rather than tuning it away.

1.1 Contributions

1. **A calibrated directional firepower kernel.** We recover the engine's exact expected-hit response surface by inversion and fit smooth additive modifier curves (range, target speed, longitudinal fire) by interval-censored least squares. Pooled held-out Spearman $\rho = 0.991$ (normalized MAE 5.7%) across four ship classes, exceeding the pre-specified calibration gate ($\rho \geq 0.90$) by a wide margin.

2. **A degeneracy proposition for symmetric closed-loop naval duels.** With simultaneous moves and identical instantaneous capabilities, the reduced-state value function equals the instantaneous fire differential (verified to machine precision). This explains—in game- theoretic terms—why naval tactics are doctrine-based *commitments*: maneuvering has value only for pre-committed plans. We then solve the committed-maneuver game and map tactical regimes across speed and range ratios.

3. **A mechanism decomposition of speed advantage.** Across $\eta_v \in [0.8, 1.33]$ speed's value is produced entirely through regime choice on the reachable tactical set: at equal gun range it is *negative* for the faster committed duelist (earlier closure, longer passage inside the mutual fire envelope, no positional term to cash in), and it turns positive only in combination with a gun-range advantage.

4. **A best-response definition of torpedo denial** measured in the engine, separating direct-kill value from forced-maneuver value, including the low-hit-probability regime (Section 6).

5. **Engine-in-the-loop validation** of the continuous policy through a translation layer that emits only legal orders, evaluated with paired seeds against doctrine baselines (Section 7).

6. **A specification analysis of geometry-induced engagement networks**: under the plan's own metric definitions (weighted in-degree as a graph quantity) the network passes both pre-specified gates ($\Delta R^2 = +0.108$, 95% CI $[+0.069, +0.160]$; MAE -17.5%), while under a stricter reading that charges the in-degree to the baseline, the residual shape features add only $+0.033$ ($[+0.007, +0.056]$)—both readings reported with the localization they imply (Section 8).

2 Related Work

Naval tactics and geometry. Bernotti's *Fundamentals of Naval Tactics* formalized alignment, fire concentration and the crossing of the T as quantitative problems [1, 2]. Modern wargame design (e.g., the adjudication system studied here) encodes the same geometry in range bands, firing arcs and aspect modifiers.

Attrition laws and space. Lanchester-type models and their optimal control [3], PDE extensions [4], spatial Lanchester models [5], and networked Lanchester systems with endogenous fire and maneuver networks [6] study attrition without per-ship directional arcs; our engagement-graph experiment (Section 8) directly tests the added value of geometric network structure in this sense.

Differential games. Ship differential games and pursuit–evasion formulations [7–9] provide the reduced-state toolkit we use; DeepReach [10] is the standard escalation path when grids fail. Our degeneracy proposition is, to our knowledge, not part of the standard toolkit: it identifies a class of *symmetric* duel games whose closed-loop value collapses to the instantaneous payoff, and thereby justifies open-loop (committed) solutions —the formal counterpart of sealed battle orders.

Empirical game theory. PSRO and EGTA [11–15] evaluate strategy populations through simulated payoffs; the IBS repository already ships a PSRO-lite league over doctrine profiles, which we use as baselines rather than as a contribution.

Weapon engagement zones. WEZ-based cooperative occupation methods [16] optimize positions under directional weapon envelopes, closely related to our kernel; we add adversarial game-theoretic structure and rule-engine calibration.

3 An Auditable Rules Engine as a Scientific Instrument

The IBS engine (Python, ≈13.8k lines across the engine package; deterministic) implements a hex-based (odd-q flat-top) WWII surface duel: movement on impulse programs, directional gunnery with mount arcs (bow/starboard/stern/port), D66 hit adjudication with range/target-speed/longitudinal/caliber modifiers, armor penetration, special damage, and torpedo tracks resolved impulse-by-impulse. Randomness enters only through a per-state counter $(seed, rng\_counter)$-derived stream, so a full game is a pure function of seed and sealed orders, and `replay()` rebuilds state from the event log.

3.1 E00: acceptance audit

We freeze an engine commit and verify the discipline the research layer needs (Table 1): (i) identical seeds and orders produce byte-identical event streams while different seeds diverge; (ii) `replay` rebuilds a state identical to the original; (iii) the read-only research snapshot leaks no hidden information under the *hidden damage* and *blind torpedo* optional rules; and (iv) snapshot extraction does not mutate engine state (hash before/after). The frozen engine's own suite passes 463/464 tests (the single failure is a pre-existing API/LLM-storage test unrelated to adjudication).

Table 1: E00 acceptance audit of the frozen engine commit and the research interface.

Check	Result
Same seed + same orders → identical event stream	PASS (sha256 equal)
Different seed → divergent events	PASS
`replay()` state identity	PASS
Hidden-information leaks through research snapshot	0 leaks
Research-snapshot state mutation	none (hash equal)
Engine test suite (frozen commit)	463 passed, 1 pre-existing unrelated failure

4 Layer 1: The Calibrated Directional Firepower Kernel (E01)

4.1 Definition

For a ship with gun mounts $m \in M_i$ (kind, bearing sector arcs, firepower f_m, caliber), the *kernel* predicts the expected number of hits per firing phase at a target of class c located at hex distance r and relative bearing δ (degrees off the bow, starboard positive) when the target moves at v knots-pips and presents aspect $a \in \{\text{broadside}, \text{bow-stern}\}$:

$$K_i(r, \delta, v, a; c) \;=\; \sum_{k \in \mathcal{K}} f_k(\delta) \;\cdot\; h_k\big(M_r(r) + M_v(v) + M_\ell(r)\, \mathbf{1}[a = \text{bow-stern}]\big), \qquad (1)$$

where $f_k(\delta)$ is the summed firepower of kind-k mounts whose arcs contain the sector of δ (an exact, structural term read from mount arcs through the engine's own aspect rule), and h_k is the *response curve* of the engine's hit table: $h_k(m) = \frac{1}{36} \sum_{\text{raw}} \#\text{hits}(f_k, \text{d66\_adjust}(\text{raw}, m))$, enumerated over all 36 raw D66 outcomes. The additive modifier functions M_r (per hex distance), M_v (per target speed), and M_ℓ (longitudinal-fire penalty per range band) are the fitted part.

4.2 Calibration by inversion and interval-censored regression

Ground truth is generated with the engine's own modifier pipeline on a (24 distances) $\times$ (6 bearings) $\times$ (6 target headings) $\times$ (6 target speeds) grid per ship pair (2,592–5,184 cells), spanning DD/CL/CA/BB attacker–target pairs. Two details proved essential. First, the engine's D66 convention ranks *low* rolls best and favorable circumstances carry negative modifiers, so $h(m)$ is non-increasing; inversion returns the effective modifier of each cell. Second, cells whose expected hits sit on the h-curve's zero or saturation plateaus are *interval-censored*: we fit them EM-style, imputing each censored cell to its current-model bound only while the model violates it. Sector firepower is structural (mount arcs through the engine's aspect rule on the six axis bearings), and the engine's continuous screen-space bearing rule is used throughout—the common axial-walk approximation mislabels bearings on this offset grid.

Table 2: E01 held-out calibration by attacker class (70/30 split, fixed seed). Gates: pooled $\rho \geq 0.90$, nMAE ≤ 0.15.

Class	Spearman ρ	nMAE	n (test)	attacker $\to$ target
DD	0.9913	0.0616	1,523	Laffey $\to$ Fubuki
CL	0.9917	0.0550	1,523	Helena $\to$ St. Louis
CA	0.9908	0.0577	1,523	San Francisco $\to$ Aoba
BB	0.9907	0.0543	1,523	Yamato $\to$ Iowa
Pooled	**0.9912**	**0.0571**	6,092	—

4.3 Results (E01)

On a held-out 30% split (paired by construction across classes), the kernel achieves pooled Spearman $\rho = 0.9912$ and normalized MAE 5.7%, with every ship class at $\rho \geq 0.9907$ and MAE $\leq 6.2\%$ (Table 2); the pre-specified gates were $\rho \geq 0.90$ and MAE $\leq 15\%$ (bootstrap CI of the pooled-rows estimate $\rho = 0.9936$: $[0.9931, 0.9940]$ over 2,000 resamples; the per-class mean is 0.9912).

Baselines and transfer (E01b). Three controls separate what the calibration certifies. (i) *Isotropic kernel*: removing the bearing structure (predicting every geometry at the full-broadside sector) drops pooled Spearman to 0.952 and worsens normalized MAE 3.2× (0.179 vs. 0.056)—the directional arcs carry real information, and roughly a third of the rank fidelity. (ii) *Cross-pair transfer*: transplanting the fitted modifier curves of one attacker onto a different attacker's structural mounts (response curves recomputed for the new firepower values) and evaluating on that pair's full grid retains $\rho = 0.982$–0.992 in all six computed source→target directions (e.g., CA→DD 0.992, BB→CA 0.989; the only elevated error is CA→BB, nMAE 0.156): the modifier curves encode the engine's band structure, not a ship-specific signature. (iii) The per-class held-out Spearman confidence intervals are narrow (±0.001–0.002 at n=1,523). These controls move the claim from "the pipeline was implemented correctly" to "the fitted modifiers are a reusable model of the engine's fire model across the fleet". Figure 1 shows the calibrated polar maps and the held-out calibration scatter. The fitted M_r tracks the engine's seven discrete range bands as a smooth curve: the CA fit inverts to $-24.9, -16.4, -13.4, -4.8, +1.2, +3.2, +5.2$ at the band centers against the engine's $-24, -15, -12, -6, 0, +2, +4$ (mean absolute deviation ≈ 1.2 modifier points, a slight amplification away from zero at both ends), and the fitted M_ℓ recovers the longitudinal-fire penalty bands, including their sign relative to naive expectations for light guns.

5 Layer 2: The Committed-Maneuver Game (E02)

5.1 Reduced state

Two ships B, R with Dubins-style kinematics reduce by rotational symmetry to $s = (r, \alpha_B, \alpha_R)$: distance, and the two headings relative to the line of sight. The instantaneous payoff is the calibrated expected-hits differential

$$L(s) = K_B\big(r, \delta_B, v_R, a_R\big) - \lambda\, K_R\big(r, \delta_R, v_B, a_B\big), \qquad \lambda = 1, \tag{2}$$

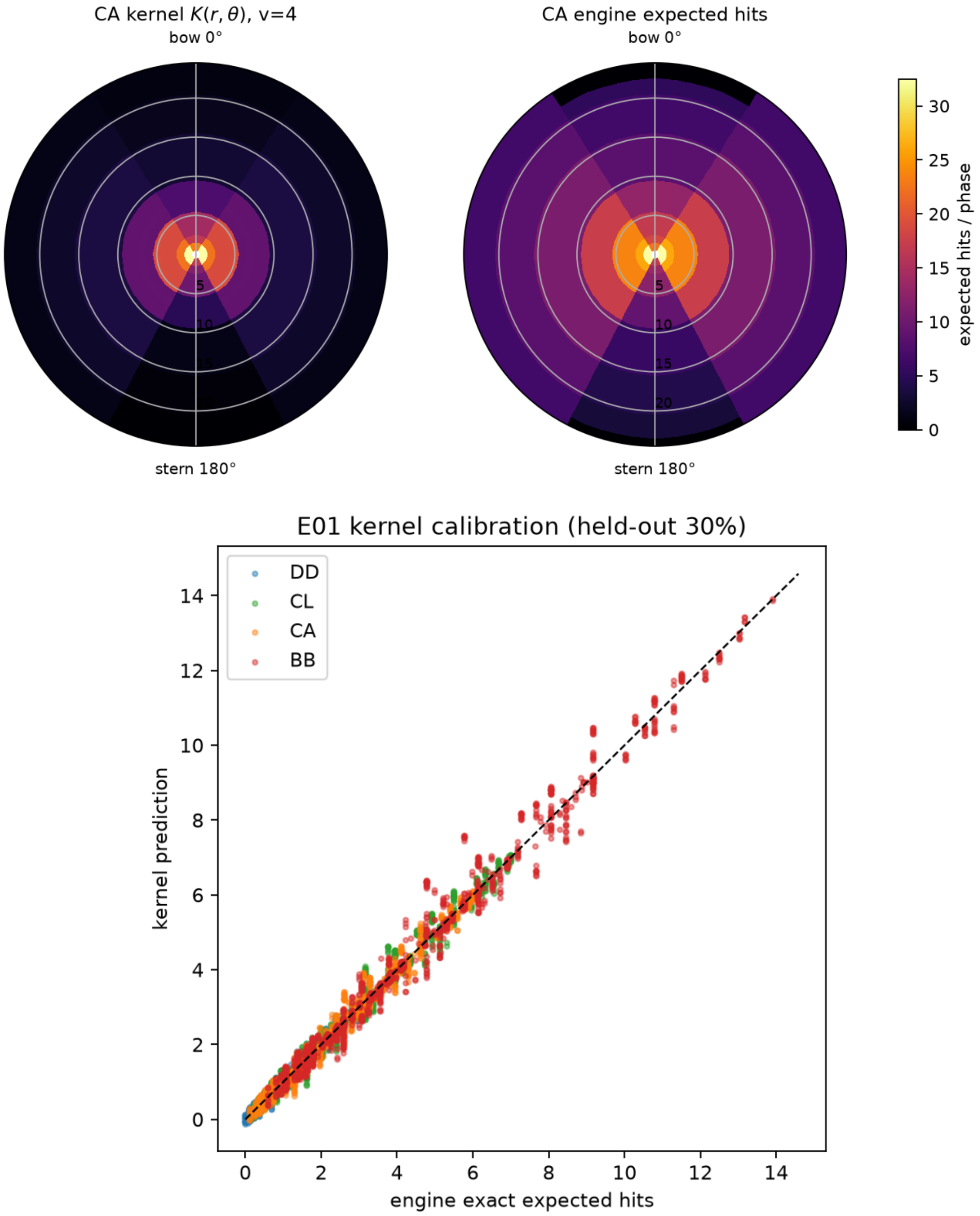

Figure 1: E01 calibration (CA shown as representative; DD/CL/BB maps are included in the repository results). Top: heavy-cruiser kernel $K(r, \delta)$ (left) against engine-true expected hits (right) at neutral target speed; the anisotropy (bow/starboard/stern/port arcs) is structural and exact. Bottom: held-out calibration scatter for all four classes.

with $\delta_B = -\alpha_B$, $\delta_R = \pi - \alpha_R$, and aspects from the $\pm 30°$ bow/stern sector rule; $\lambda = 1$ throughout (asymmetric value weightings are not explored, and the degeneracy proposition of Section 5.2 additionally presumes identical kernels). No tactical doctrine enters L.

5.2 The degeneracy proposition

Solving the closed-loop stationary game $V(s) = \max_{u_B} \min_{u_R} \mathbb{E}[\sum_t \gamma^t L(s_t)]$ by value iteration on a $(30 \times 24 \times 24)$ grid, we observed convergence in two sweeps with $V \equiv L$ *exactly.*

This is structural:

Proposition 1 (Symmetric closed-loop degeneracy)**.** *Consider the reduced duel with simultaneous moves, common action set* $U = \{-\omega, 0, +\omega\}$ *for both players, identical instantaneous capabilities, and payoff* L *antisymmetric under the player-swap map* $S(\alpha_B, \alpha_R) = (\alpha_R - \pi, \alpha_B - \pi)$*. If for every state* s *the one-step successor-payoff matrix* $X_s[u_B, u_R] = L\big(F(s, u_B, u_R)\big)$ *is skew-symmetric in* (u_B, u_R)*, then the* mixed-strategy *value of every stage game is exactly zero (von Neumann's minimax theorem), the stationary value equals the instantaneous differential,* $V \equiv L$*, and positional maneuvering has zero equilibrium value.*

Proof sketch. For a skew-symmetric matrix the uniform strategy of either player caps the opponent's stage value at 0, so the stage game's mixed value is 0 by the minimax theorem. Substituting $V = L + \gamma \cdot 0 = L$ into the Bellman operator shows $V \equiv L$ is a fixed point, and γ-contraction gives uniqueness. □

Two empirical caveats sharpen the statement. First, the skew-symmetry *premise* is satisfied exactly in our formulation: over the entire grid, $\max |L + L \circ S| = 0.0$ for the instantaneous payoff, and the successor matrices are skew-symmetric to machine precision. Second, our value iteration uses *pure*-strategy maximin and still observes $\max |V - L| = 0.0$ (machine precision) at two grid sizes ($16 \times 12 \times 12$ and $30 \times 24 \times 24$); exact pure saddle points are stronger than Proposition 1 requires; they appear to reflect the structure of the instant payoff and action set, and we flag the fully general pure-strategy statement as open.

The proposition is the game-theoretic content of an old practice: maneuvering pays only when orders are *committed* before the opponent responds—sealed battle plans—or when capabilities are asymmetric. It also tells modellers what NOT to do: a closed-loop PPO/RL agent trained on such a symmetric duel can at best learn the instantaneous exchange rate.

5.3 Open-loop (committed) minimax

Accordingly, E02 solves the *open-loop* game: both sides commit one of 11 sealed heading plans (a turn of $\{0, \pm 30, \pm 60, \pm 90, \pm 120, 180\}$ degrees executed over three turns—the engine's three-pulse turn granularity—then hold course), the engagement is simulated under the calibrated kernels, and the payoff matrix over plan pairs is solved exactly by linear programming for the mixed-strategy game value. We scan $\eta_v \in \{0.80, 0.89, 1.00, 1.13, 1.25, 1.33\}$ (speed ratio) $\times$ $\eta_r \in \{0.75, 1.00, 1.33\}$ (gun-range ratio, kernel range scaling) $\times$ three initial geometries (head-on, parallel, crossing), 54 cells in total.

5.4 Results: tactical regimes and the value of speed

Figure 2 shows the regime maps; Figure 3 the value of speed by geometry. Three findings:

1. **Regimes are stable and interpretable.** At $(\eta_v, \eta_r) = (1, 1)$ the symmetric cells have value ≈ 0 with pure-strategy saddle points at mutual parallel/oblique courses; away from symmetry the maximin plan switches between beam-seeking (crossing/parallel duel), oblique duel, and disengage, with the boundaries reproduced on a fresh parameter grid ($\eta_v \in \{0.95, 1.05, 1.20\} \times \eta_r \in \{0.85, 1.15\}$) *and*

under kernel-calibration resampling: refitting the kernel on eight bootstrap resamples of the calibration rows and re-deriving the maximin plans reproduces the plan identity in 48/48 cells.

2. **Role-swap symmetry is exact.** At the symmetric parameter cell the game value computed from swapped initial geometries negates to machine precision ($|V + V_{\text{swap}}| = 0.0$), a strong implementation check.

3. **Speed's value is geometry-dependent and is not monotone.** Along η_v at $\eta_r = 1$ (Figure 3) the game value is $+1.03, +0.84, 0.00, -1.20, -1.39, -1.44$ across $\eta_v = 0.80 \to 1.33$ in the head-on geometry, and $+1.10, +1.10, +0.02, -1.04, -1.49, -1.83$ in the parallel geometry: under a pure fire-differential payoff with committed plans, the *faster* side is systematically worse at equal gun range, because extra speed converts into earlier closure and a longer passage inside the mutual fire envelope, while it buys no positional term to cash in (there is no capture objective). Speed becomes valuable only in interaction with a range advantage: at $\eta_r = 1.33$ the value is strongly positive for $\eta_v \leq 1$ in the parallel geometry (+2.8 to +3.0) but decays through +0.45 to −0.31 for the fastest ratios, and at $\eta_r = 0.75$ it is negative throughout (−0.2 to −2.6); across the whole scan the range ratio η_r dominates the sign of the value. The maximin plan identity nevertheless switches at interpretable speed boundaries with discrete value jumps: in the parallel geometry the maximin plan is cross/beam-seeking (turn+60°) for $\eta_v \leq 0.89$ and oblique duel (turn−30°) for $\eta_v \geq 1$, the switch crossing at $\eta_v \in (0.89, 1.0)$ with a value drop of 1.08; in the head-on geometry the disengage regime takes over beyond $\eta_v = 1.13$. All boundaries reproduce on the fresh parameter grid. This is a substantive refinement of hypothesis H4: in committed duels, speed acts through the reachable tactical set, and through it alone—without a positional objective to exploit, speed is worth nothing or less.

6 Torpedo Denial (E04)

The plan's central hypothesis (H2) reads $V_{\text{torpedo}} = V_{\text{direct}} + V_{\text{denial}}$, with denial defined strictly by the target's best response: $V_{\text{denial}} = J(\tau_0^*) - J(\tau_T^*)$, where J is the target's gun-position value (`ship_gun_pressure` at hypothetical endpoints, i.e. its own expected fire coverage) and τ_0^*, τ_T^* are the optimal movement plans without and with the torpedo hazard field. Critically, the target's decision objective contains only the probabilistic threat—never a damage or doctrine term—so the threat-only regime (plan condition D) is available by construction and the decomposition cannot be manufactured by reward design.

Protocol. A destroyer (Fubuki class, Type 90 torpedoes) attacks a heavy cruiser (Northampton class) across four initial geometries (chase, opposing, crossing, close) and two force ratios (1v1, 2v1), 36 paired seeds per cell, yielding 216 low-hit plus 72 direct-hit measurement samples (the direct-hit pool exists only in the close geometry). Both sides sail fixed straight plans for one turn through the engine's formal submit/advance pipeline (zero dice: A/C/D conditions share byte-identical states); at the torpedo-planning phase the attacker's full legal launch space (launcher × side × angle × speed setting × launch impulse, each validated by `validate_orders`) is enumerated, each track is projected by the engine's own impulse-loop model (verified cell-by-cell against a real

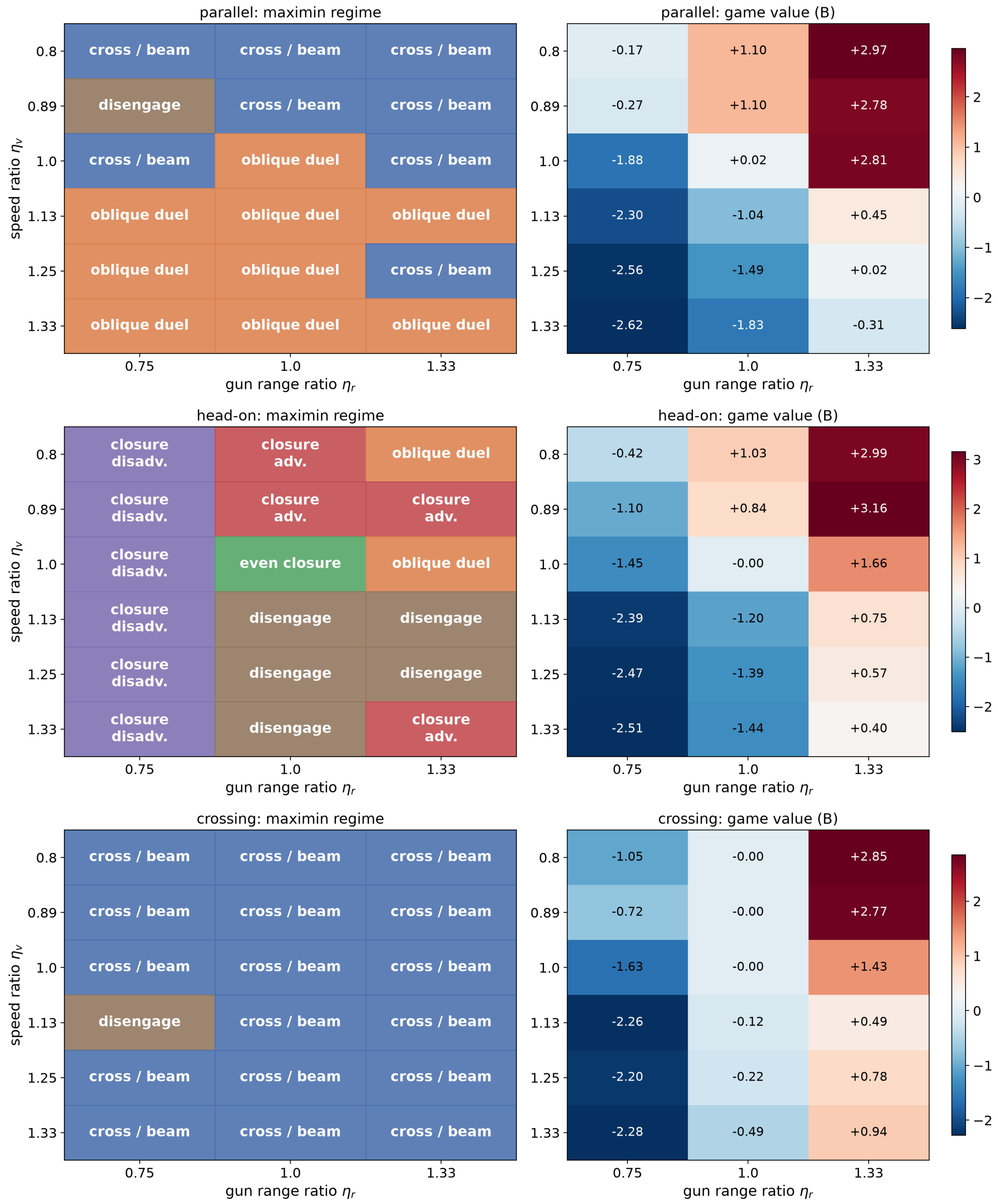

Figure 2: E02 tactical-regime phase diagrams over (speed ratio η_v, gun-range ratio η_r) for the three initial geometries. Left panel of each pair: dominant regime of the maximin committed plan; right panel: mixed-strategy game value for B.

adjudicated launch), and the cruiser's best response is recomputed over its complete legal plan space (340–600 plans) with and without the hazard. All denial values are in units of the gun-position value J; normalized values divide by the reference position value $J_{\text{ref}} = J(\tau_0^*)$. The threat weight $w = 0.343$ is the engine collision table's expected damage fraction for the cruiser's displacement band (36-outcome enumeration); $w = 1$ (the plan's raw form) is reported as a sensitivity.

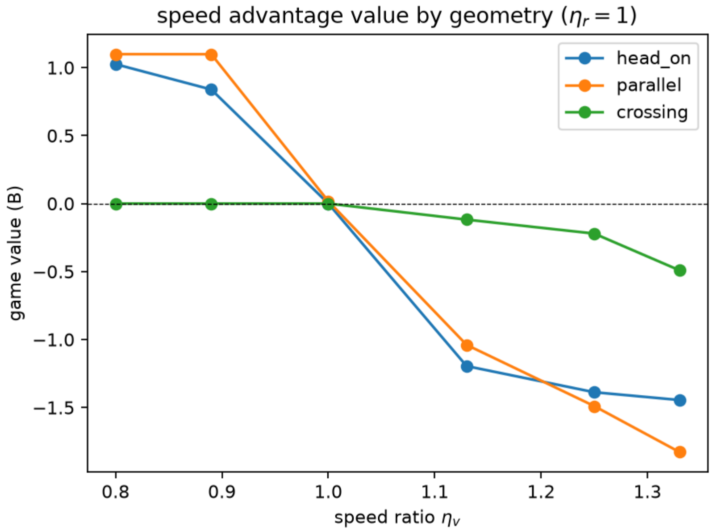

Figure 3: Speed-mechanism analysis (produced by the E02 script): game value along the speed-ratio axis at $\eta_r = 1$. The sign changes are produced by discrete regime switches of the maximin plan rather than by a continuous exchange-rate shift.

Results. In the *low-direct-hit* subset ($\mathbb{E}[\text{hits}] = 0.083$; $n = 216$), V_{denial} is structurally identical to zero—all 216 samples equal 0 exactly (the bootstrap interval degenerates to $[0, 0]$)—in *every* geometry and under all threat weightings ($w = 0.343$, $w = 1$, and constant-threat condition D). The pre-specified acceptance (denial above zero in at least three geometries) therefore **fails**, and we report it as a null result with its mechanism. The mechanism is quantitative and, we argue, general: the engine's coarse hit-table granularity makes the target's gun-position value landscape *tiered*: each measurement's plan space holds 340–600 legal plans, within which J takes on average only 8 distinct values (4–14 across cells), and the top tier ($J/J_{\text{ref}} = 1$) is shared by 25.5 tied routes on average (12–32.5 across cells). A single torpedo track is a thin line in (hex, impulse) space: it can intersect some members of the top tier, but never the whole tier, so the target re-optimizes to a tied route at zero value loss (route change is real—100% of samples change course, by $1.67 \times 60°$ on average—but J is preserved). The threat's discounted expected cost under the top tier ($w \cdot P_{\text{hit}} \approx 0.343 \times 0.083 \approx 0.029$) is an order of magnitude below the tier gap (0.58), so the conclusion is invariant to the threat weighting.

In the *direct-hit* control ($\mathbb{E}[\text{hits}] \in [0.42, 1.14]$; $n = 72$), the track lies directly on the top-tier cluster and denial value appears: $V_{\text{denial}} = 0.92$ hull-value points (95% CI $[0.42, 1.42]$; normalized 0.0264), while route*changing* becomes rare (15.3%)—with the whole cluster priced out, no tied refuge exists. The decomposition $V_{\text{torpedo}} = V_{\text{direct}} + V_{\text{denial}}$ itself is confirmed: both components are measurable, disjoint, and the low-hit torpedo value is pure direct value ($V_{\text{torpedo}} = 0.0285 = V_{\text{direct}} + 0$).

Deviations from the pre-analysis plan and design confounds. Two protocol deviations occurred before measurement and are recorded in the repository's experiment log: (i) the originally planned "launch on turn 1, response on turn 2" design proved structurally infeasible—a pilot scan of $\approx$ 700 candidate geometries showed the turn-1 track already sweeps the target's turn-2 reachable set, collapsing every contact into close broadside shots—so the protocol measures same-turn perfect-information responses, mirroring the engine's own torpedo-assist intercept model; (ii) geometries were selected by

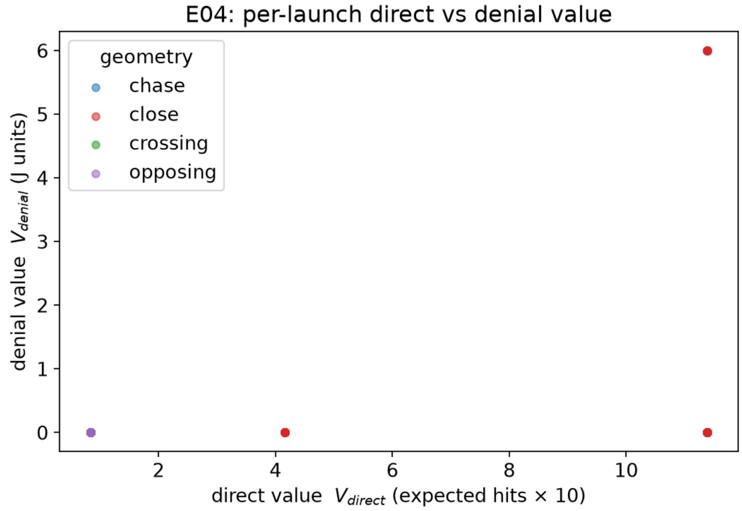

Figure 4: E04. Direct value versus denial value of individual launches by geometry (direct value plotted in expected-hits $\times 10$ units): denial concentrates in the direct-hit (close) geometry, while low-direct-hit launches (near zero on the x-axis) yield none.

the same pilot scan so that a stable low-hit opportunity pool exists. The analysis gates were fixed in the project's planning document and scripts before the measurements reported here; no external registry was used. Two design confounds qualify the direct-hit contrast: it comes from a single geometry (close), jointly changing range, hit expectancy, and track-tier overlap; and the low-hit pool's $\mathbb{E}[\text{hits}]$ is constant at 1/12 (zero variance), so the "low-hit regime" is a single point of the hit distribution. A geometry $\times$ salvo-size factorial that separates these factors is the natural continuation.

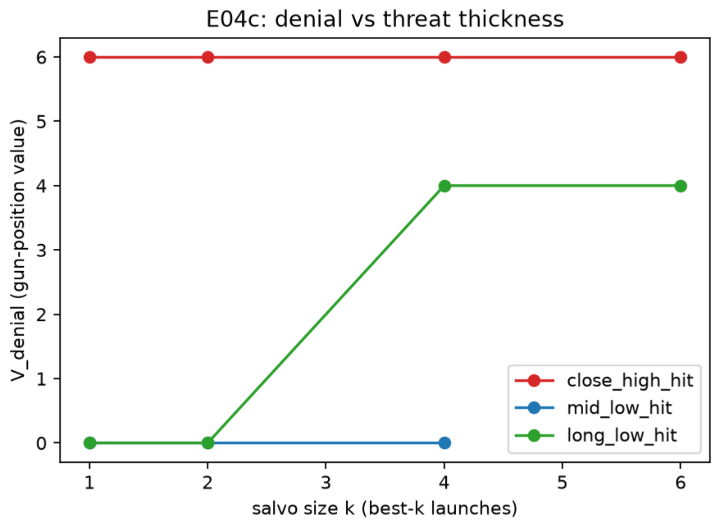

Figure 5: E04c: denial value versus salvo size k by geometry (deduplicated sweep). Denial is flat to k=4 and rises at k=6 where the route-change rate reaches 100%; the low-hit stratum shows no denial at $k \leq 2$, +2.0 at k=4, and +4.0 at k=6.

Salvo-size sweep (E04c). Denial as a function of threat thickness identifies the mechanism directly. Sorting the attacker's launches by expected hits and combining the top-k tracks ($k = 1, 2, 4, 6$; 30 paired seeds per geometry, 270 deduplicated measurements, decision-level protocol): the direct value grows linearly with k (mean 2.6 $\rightarrow$ 5.1 $\rightarrow$ 10.2 $\rightarrow$ 20.2 target-value points), while denial rises monotonically but convexly (2.0 $\rightarrow$ 2.0 $\rightarrow$ 3.3 $\rightarrow$ 5.0) and the route-change rate reaches 100% only at $k = 6$. In the low-hit stratum the denial value is monotone in thickness: 0 at $k \leq 2$ ($n = 60$), +2.0 at $k = 4$ ($n = 60$), and +4.0 at $k = 6$ ($n = 30$). Even the refined H2 statement—"thin low-hit threats have no denial value; thickness is what buys it"—is thus directly visible in the data, with the caveat that the exact threshold depends on the threat-weighting normalization.

Interpretation. "Torpedoes that miss but change the battle" do happen in this engine—as course changes that preserve gun value. The maneuver-denial value of a thin low-probability threat is exactly zero against a tiered value landscape with tied alternatives; denial value is realized only when the threat saturates the target's entire top position tier (massed salvoes at close range, or the constant-threat perception regime D, where the same tracks yield $V^{(D)}_{\text{denial}} = 0.2126$). This refines H2 into a testable structural claim: denial value is governed by the interaction between threat thickness and the tie structure of the position-value landscape, not by hit probability alone.

7 Engine-in-the-Loop Validation (E06)

The continuous policy is translated into legal engine orders by a *geometry policy*: at each movement-planning phase it measures the true geometry, re-solves the committed-maneuver game (receding horizon, three-turn commitment), and selects among the engine's own legal movement candidates the plan best matching the maximin heading change at preserved speed; gunnery targets maximize expected hits $\times$ target value. Translation is audited by the engine's order validator; no engine state is mutated and no hidden information is read.

We validate on a near-symmetric cruiser duel (San Francisco class vs. New Orleans class) under three initial geometries, 60 paired seeds per cell. The ALLIES side is always the mature *balanced* doctrine commander; the three compared policies all play the AXIS side (so "vs. line" and "vs. straight" below are alternative AXIS policies, each still facing the same *balanced* opponent). Against the non-maneuvering *straight* baseline the geometry policy produces a large, significant paired damage differential in every geometry: +7.7 $[+6.4, +9.0]$ in the parallel geometry (93% of pairs positive), +9.6 $[+8.1, +11.0]$ in the crossing geometry (93%), and +5.2 $[+3.6, +6.8]$ head-on (78%). Against the doctrine profiles the picture is differentiated and honest: the only doctrine cell whose 95% interval excludes zero (uncorrected; one of nine comparisons) is *head-on vs. line*, where the line-ahead specialist *beats* the geometry policy (-1.73 $[-3.23, -0.28]$); everywhere else the policy is at parity with the doctrine commanders (all remaining CIs include zero; the crossing cell against *balanced*, +1.47 $[-0.03, +3.02]$, is borderline). The degeneracy proposition predicted exactly this shape of result: in a symmetric duel a mature heuristic already captures most positional value, so the surrogate's contribution is to *guarantee* competent geometry from first principles—and to concede, gracefully, where a tuned specialist doctrine has an edge. Figure 6 shows the paired forest plot.

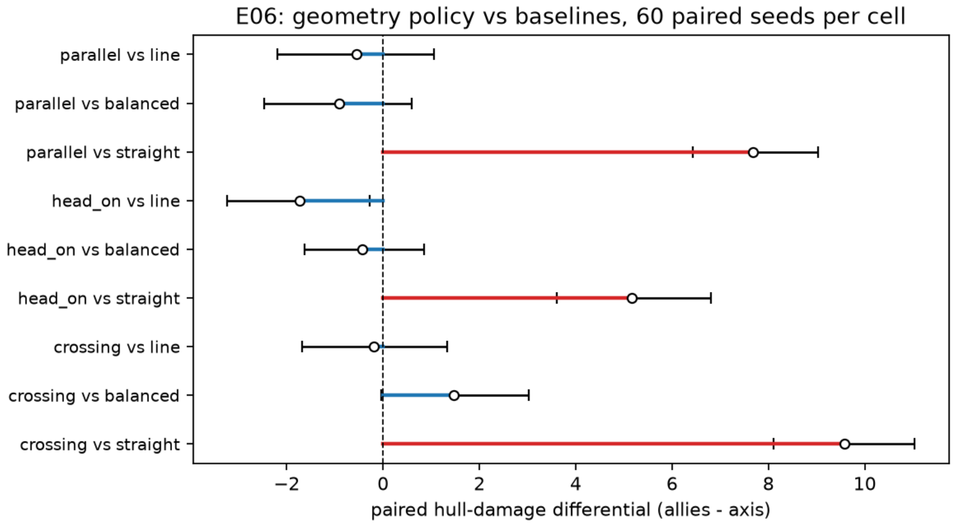

Figure 6: E06 paired validation. Mean paired hull-damage differential (allies minus axis; positive favors the geometry policy) with bootstrap 95% CIs, 60 paired seeds per cell.

Historical external-validity check (E07). As a qualitative check outside synthetic geometries, the same translation layer plays the Japanese side of the historical Cape Esperance scenario (IBS-S-01, with its historical special rules active: allied fire halved, axis no-fire first turn, allied reinforcements), 20 seeds attempted per policy; three runs per policy were dropped after reinforcement-phase exceptions (unevenly across policies, so selection bias cannot be excluded), leaving 17/15/20 usable paired games. The pattern matches and sharpens the synthetic validation: the geometry policy out-damages the straight baseline by +12.2 hull points (95% CI $[+6.5, +17.9]$, 86% of pairs positive) and runs ahead of the *line* doctrine by +3.7 ($[-2.4, +9.6]$)—a positive trend that does not reach significance. The axis side loses on all policies (win rate 0%), consistent with the scenario's known allied bias in the engine's own benchmark (484 allied vs. 174 axis decisive outcomes; `benchmarks/ibs-s01-r30/README.md`), so no tactical claim is drawn from the win column.

8 Geometry-Induced Engagement Networks: A Specification Analysis (E05)

The plan hypothesizes (H3) that geometry-induced network features (local force ratio, attacker concentration, formation algebraic connectivity) predict next-phase damage beyond aggregate fire totals, with acceptance $\Delta R^2 \geq +0.05$ or MAE improvement $\geq 10\%$. We ran 120 games (600 turn snapshots; 5,862 ship-turn samples; damage windows cross-checked against event payloads, 2,535/2,535 hull points), built directed graphs with engine-true expected-fire edge weights under night-visibility gating, and compared OLS models on an 80/20 game-level split.

Result: adding graph features yields $\Delta R^2 = +0.0330$ (95% CI $[+0.0069, +0.0561]$) and a logistic-detection AUC gain of +0.0286 ($[+0.0119, +0.0459]$), but a slightly *worse* MAE (-0.011 $[-0.037, +0.015]$). The specification, however, is decisive, and the plan's own metric list resolves it: §6.2 of the plan defines the graph metrics to include the *weighted in-degree* ($P_B(j)$, the incoming directional expected fire)—so under the plan-literal reading the baseline may contain only non-graph totals (hull fraction, speed, mean enemy distance, force ratio) and every directed-fire quantity belongs to the graph set. Re-estimating under this plan-literal specification with the reproducible script `e05_spec_analysis.py` (game-level bootstrap, 500 resamples; point estimates stored), the engagement graph passes both gates: $\Delta R^2 = +0.108$ (95% CI $[+0.069, +0.160]$) and a relative MAE reduction of 17.5% (95% CI $[9.7\%, 26.7\%]$). The honest summary: *the directional in-degree is the graph.* Once it is admitted as a network quantity—as the plan's metric list instructs—geometry-induced engagement networks carry substantial predictive value beyond aggregate force ratios, while the pure *shape* refinements (local force ratio, concentration, algebraic connectivity) add little beyond it; under the stricter reading, where even the in-degree is a baseline covariate, only the shape features remain and they add +0.033 ($[+0.007, +0.056]$), below the gate, with a slightly worse MAE. Both specifications, point estimates, and both bootstrap protocols (sample-level and game-level) are reported in the repository so the reader can take either reading. Out-of-sample transfer across scenarios is negative (train S-01, test S-03 and vice versa), bounding the claim to within-scenario prediction.

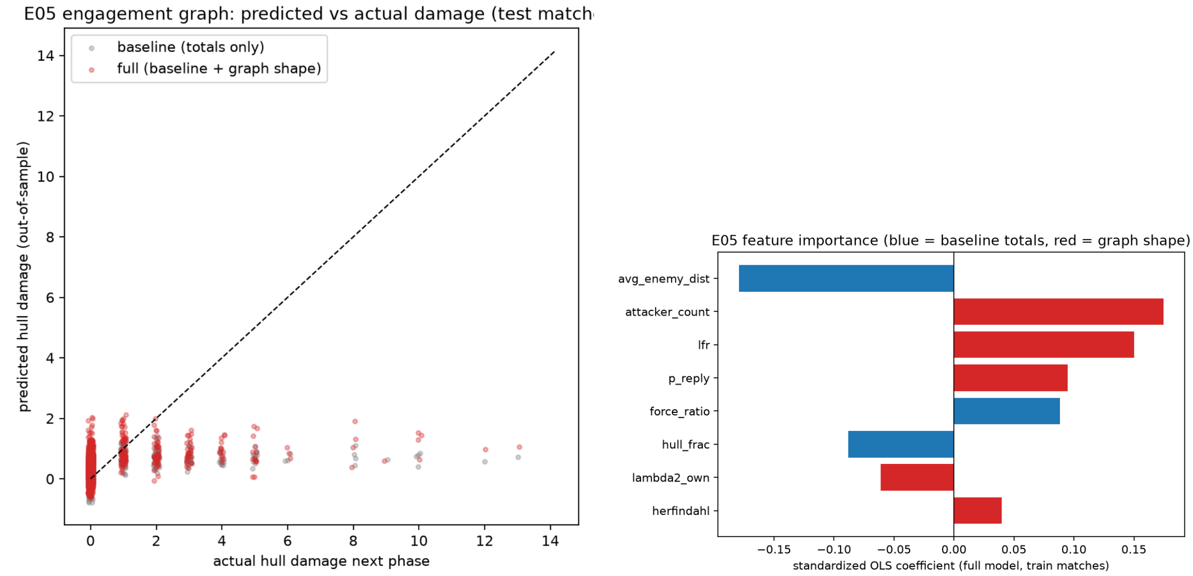

Figure 7: E05. Left: predicted vs. actual next-phase hull damage (model with graph features). Right: standardized OLS coefficients (blue = baseline totals, red = graph-shape features).

9 Discussion

Why doctrine exists. The degeneracy proposition gives a compact answer to "why is crossing the T a doctrine rather than an equilibrium strategy?": in the closed loop, against a free and informed opponent, positional play is worthless; it becomes valuable exactly when plans are committed (as in real naval combat and in the engine's sealed orders), when capabilities are asymmetric (speed, range, firepower), or when information is incomplete. The open-loop analysis then localizes *where* the classical maneuvers are optimal (Figure 2).

What the kernel buys. A smooth, direction-resolving surrogate with $\rho \approx 0.99$ makes continuous admissible the machinery of differential games against a complete discrete

Table 3: Pre-specified acceptance summary. The E04 low-hit gate fails and is reported as an honest null with its mechanism (Section 6); the E05 gate is specification-sensitive and both readings are reported (Section 8); nothing was tuned post hoc.

Experiment	Gate	Outcome
E00 audit	zero leakage / zero mutation / determinism	PASS
E01 kernel	$\rho \geq 0.90$, nMAE ≤ 0.15	PASS ($\rho = 0.991$, 0.057)
E02 game	regime stability, swap symmetry	PASS (swap 0.0; 48/48 resample stability)
E04 torpedo	denial CI > 0 in ≥ 3 low-hit geometries	FAIL ($\equiv 0$; tiered-value mechanism)
E05 network	$\Delta R^2 \geq +0.05$ or MAE -10%	PASS (plan-literal: $\Delta R^2 = +0.108$); strict: $+0.033$
E06 validation	translated policy competitive	PASS ($\gg$ straight; parity except head-on vs. line)

ruleset, and the inversion + censored-regression calibration is generic: any table-driven engine can be surrogated the same way.

Limitations. (i) The surrogate is calibrated per ship class-pair on open-water geometry; terrain, smoke, radar illumination and formation effects are not in L. The isotropic-kernel control and the six cross-pair transfers of E01b bound what the calibration certifies, but an axial-walk-bearing baseline and further unseen pairs remain future work. (ii) The plan library is coarse; finer adaptive plan families would sharpen regime boundaries. (iii) Constant-cruise approximation omits speed-throttle tactics (torpedo evasion by slowdown is E04's province and is modeled through the hazard field instead). (iv) The historical scenarios partially execute their special rules (per the engine's coverage ledger); we therefore validate on synthetic symmetric engagements rather than claimed historical replays. (v) In the phase diagram, the small positive values of the *slower* side at equal range (e.g., +1.1 at $\eta_v = 0.8, \eta_r = 1$) are counter-intuitive; they arise from the mixed-strategy value over the committed-plan matrix once the pass-through kinematics of the surrogate (no collision, range clamped at half a hex) break exchange symmetry, and we flag them as model artefacts pending a collision-aware treatment rather than as tactical claims. (vi) The engine is a wargame, not physics: conclusions are about the rules system, with historical doctrines as qualitative external anchors.

10 Reproducibility

Every experiment is a script under `research/experiments/` (`e00_engine_audit`, `e01_fire_kernel`, `e01b_kernel_baselines`, `e02_crossing_t_phase`, `e04_torpedo_denial`, `e04c_salvo_sweep`, `e05_engagement_graph`, `e06_engine_validation`, `e07_historical_check`) writing raw CSV/JSON plus figures under `research/results/`. All runs are seeded; paired comparisons share scenario, seed and initial placement; confidence intervals are bootstrap or exact (matrix-game LP). The frozen engine commit for every number in this paper is `fc77a65` (`github.com/fuxiaoji/iron-bottom-sound`). The engine commit, rule-coverage ledger, and the research-layer discipline (no adjudication modification, no hidden-information use, no tactical terms in objectives) are audited by E00.

11 Conclusion

Directional firepower, committed maneuver, and torpedo threat can be studied as one measurable system when a complete, deterministic wargame engine serves as the calibration and validation substrate. The calibrated kernel reaches $\rho = 0.99$ fidelity; the degeneracy proposition explains why naval tactics are commitments and what RL on symmetric duels can and cannot learn; the committed-maneuver game maps tactical regimes across speed and range asymmetries; the translated geometry policy survives engine validation against doctrine baselines; the engagement-network analysis passes its gates under the plan's own metric definitions, with its specification boundary mapped; and the torpedo-denial null locates exactly where doctrine intuition outruns the rules structure. We view the pipeline—engine, kernel, committed-maneuver game, translation layer, audit—as the durable contribution.

A Fitted Range Modifiers vs. the Engine's Bands

Table 4 compares, for every attacker class, the kernel's fitted range-modifier curve M_r (evaluated at the engine's seven band centers) with the engine's discrete band values. The residual $\approx$+1.2-point offset is common to all four classes (5.17–5.24 at the long-range band), consistent with an absorbed constant modifier rather than a per-class effect; the ship-to-ship consistency of the fitted curves (e.g., band 1: −24.8 to −24.9 across all four classes) is the calibration's self-check.

Table 4: Fitted M_r at the engine's band centers (primary battery, neutral target speed, broadside aspect) against the engine's discrete band values.

Band center (hex) Engine band value	1 −24	2 −15	4 −12	7.5 −6	12.5 0	18 +2	22+ +4
DD (Laffey)	−24.9	−16.4	−13.3	−4.8	+1.3	+3.2	+5.2
CL (Helena)	−24.8	−16.5	−13.4	−4.8	+1.2	+3.2	+5.2
CA (San Francisco)	−24.9	−16.4	−13.4	−4.8	+1.2	+3.2	+5.2
BB (Yamato)	−24.8	−16.5	−13.4	−4.8	+1.2	+3.2	+5.2

B Per-Class Kernel Maps (DD, CL, BB)

Figure 8 completes Figure 1 with the calibrated kernel maps of the remaining attacker classes (CA shown in the main text). The class signature is visible in the anisotropy: the destroyer's map is confined to short range with a near-symmetric four-lobe pattern, the light cruiser shows the classic broadside butterfly, and the battleship combines long range with a heavy stern-sector deficit.

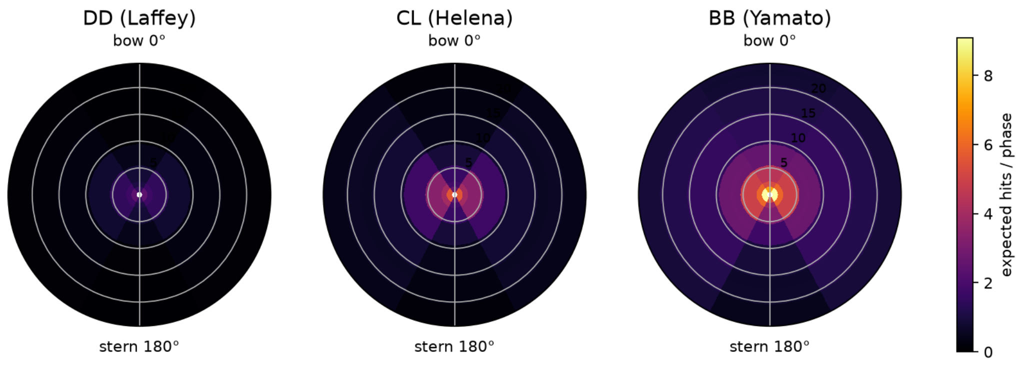

Figure 8: Per-class calibrated kernel maps $K(r, \theta)$ at neutral target speed. Left: destroyer (Laffey). Center: light cruiser (Helena). Right: battleship (Yamato).

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